SMART INDIA HACKATHON 2026

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- Problem statement ID -- SIH26264
- Problem statement Title -- Integrated dashboard for monitoring and tracking institutions related complaint cases across AICTE Bureaus/cells
- Theme -- AI POWERD SMART GOVERNANCE
- Team ID -- 197631
- Team name -- VERINEX AI



AI-Assisted case coordination & overlap Detection system

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Problem

Complaints about the same institution reach different bureaus (RB, PGRC…). Two open cases on a related issue run in parallel, so teams repeat verification, responses get delayed and officers get confused.

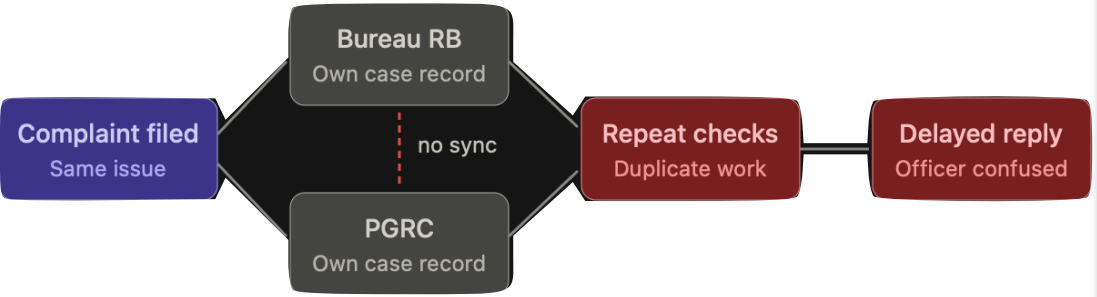
Duplicate work

Delays

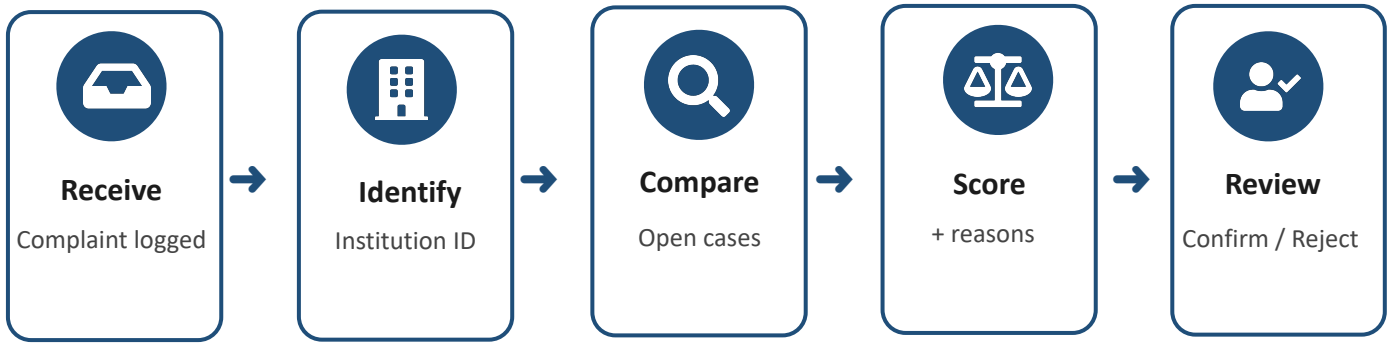
Fragmented

Current System Gaps:

- No institution-wise view of all cases
- No cross-bureau overlap alerts
- Manual tracking of status and due dates
- Decisions and reasons not recorded



Proposed Solution



Coordinate action → update case status → close. Final decision stays with the officer.

How We Stand Out

- **Explainable AI:** every alert shows a score and plain-language reasons.
- **Human-in-the-loop:** AI suggestion and officer decision stored separately; nothing is auto-merged or auto-closed.
- **Smart safeguards:** similar text at a different institution is never a strong match; closed cases treated differently.
- **Complete case journey:** Submitted → Assigned → Under Process → Transferred → Action Taken → Disposed, with full history.
- **Smart add-ons:** case summary, missing-info detector, delay-risk, priority and relationship graph.

TECHNICAL APPROACH

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Software & Technology Stack

- Python – application logic
- Flask – web routes & APIs
- PDF / Excel exports
- HTML / CSS / JavaScript
- vis-network – case graph →
- SQLite – local database

Overlap Checker (demo weights)

Institution ID	→	Category	→	Text similarity	→	Open, different bureaus
40%		20%		30%		10%

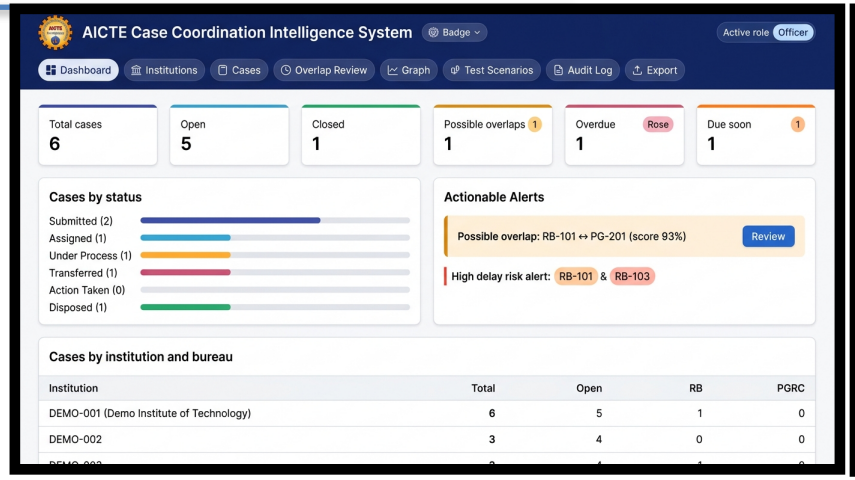
Different institution → score capped below 40, so text alone never gives a strong match. Output = score + reasons. Weights are demo values; validate before real use.

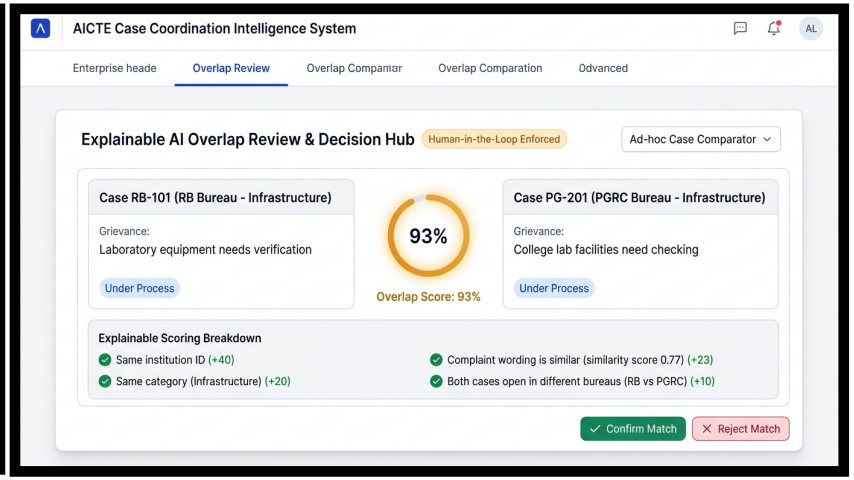
Case status journey: Submitted → Assigned → Under Process → Transferred → Action Taken → Disposed

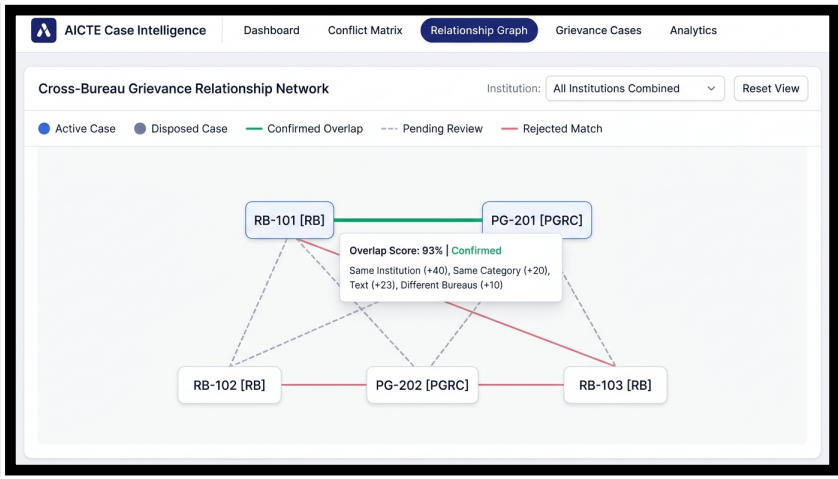
Technical Flow

Case Intake	Overlap Check	Human Review	Case Journey	Roles & Audit	Dashboard
Institution profile + case data	Score + matching reasons	Confirm / Reject with mandatory reason	Old/new status, officer, remark, time	Backend permission checks, privacy masking	Counts, alerts, reports, exports

Prototype Screens (Working Demo)







FEASIBILITY AND VIABILITY

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FEASIBILITY ANALYSIS

- Runs on Flask + SQLite; no separate server
- Zero paid services or licences
- Built in 6 phases, tested at each phase
- MVP demo works with fictional data

RISKS & CHALLENGES

- Rule-based scoring, not a trained model
- Weights and thresholds need validation
- No official AICTE data or integration yet
- Keyword match can miss paraphrased text

MITIGATION STRATEGY

- Fictional data only; privacy masking
- Officer confirms every suggestion
- Audit log of who changed what and why
- Upgrade to TF-IDF / embeddings after testing

ECONOMIC VIABILITY

- Open-source stack, zero licence cost
- Runs on an ordinary office PC
- No per-user or per-query charges
- Saves duplicate verification effort

INSTITUTIONAL VIABILITY

- Phased pilot: one bureau pair first
- Role-based permissions per officer type
- Human-in-the-loop keeps accountability
- Config-driven SLA from approved policy

LONG-TERM SUSTAINABILITY

- Move to PostgreSQL / government hosting
- Approved SLA rules replace demo values
- Retrain/tune on validated officer feedback
- Clear owner, documentation, handover

IMPACT AND BENEFITS

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1 VIEW
Institution-wise case history in one place

EXPLAINABLE
Every alert shows its matching reasons

100% OFFICER-LED
No auto-merge or auto-close by AI

Direct Stakeholder Benefits

Case Officers
See related cases early; avoid repeated work.

Bureau Heads
Coordinate across RB, PGRC and other bureaus.

Institutions
Fewer repeated queries; faster joint resolution.

Management
Dashboards for pending, overdue, escalated cases.

Quantifiable Value

Time saved
Less duplicate verification and follow-up.

Transparency
Status history and decision audit trail.

Fair decisions
Rejection reasons are recorded for review.

Reporting
Reports by institution, bureau, region, age.

DEMO, TESTING AND REFERENCES

Test Scenarios (fictional data)

A **Same institution + related lab complaints + both open**
→ Possible-overlap alert with reasons

B **Same institution, lab issue vs hostel water issue**
→ No high-confidence overlap

C **Similar lab wording, different institution ID**
→ No strong match from text alone

Also tested: rejection without reason is blocked • unauthorized roles cannot dispose cases • status history is saved • closed cases are treated differently.

Smart Features Add-on

- AI case summary
- Missing-info detector
- Delay-risk indicator
- Priority recommendation
- Duplicate history
- Relationship graph

Limitations & Honesty

- Prototype, not an official AICTE system
- Fictional data only
- No trained model; no accuracy claimed
- AI outputs are suggestions only
- SLA and due dates are demo values
- Debug mode off before deployment

Future Scope & Next Steps

- Deploy the prototype online
- Add user login with roles
- Connect a real database
- Pilot with real (approved) data